

DEUTSCHE BANK INDIA

Retail Banking — Digital Payments

REQUIREMENT TRACEABILITY MATRIX

UPI Enablement Platform

End-to-End Traceability: BRD → FRS → Use Cases → User Stories → Diagrams → UAT

Document Version: 1.0 (Draft)

Classification: Internal / Confidential

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Date: July 2026

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1. Introduction

This Requirement Traceability Matrix (RTM) provides end-to-end traceability for all 79 Functional Requirements (FR-xx) defined in the Functional Requirements Specification (FRS v1.0), linking each requirement forward to the Use Case(s), User Story/Stories, design diagram(s) (BPMN / Sequence / Activity), and UAT Test Case(s) that implement and verify it. It is the single reference for confirming that every requirement has design coverage and test coverage, and for impact-analysing any future change request.

1.1 Column Legend

Column	Meaning
FR ID	Functional Requirement identifier from the FRS
Requirement (Brief)	Short restatement of the requirement for quick reference
Pri.	MoSCoW priority (Must / Should / Could) from the BRD/FRS
Use Case	Use Case ID(s) from the Use Case Catalogue that realise this requirement
User Story	User Story ID(s) from the Product Backlog
Diagram(s)	BPMN (B#), Sequence (S#), and/or Activity (A#) diagram(s) depicting this flow
UAT TC	UAT Test Case ID(s) from the UAT Scenarios deliverable that verify this requirement

A dash (—) indicates no dedicated artefact was produced for that column, typically because the requirement is a Could Have item deferred from detailed diagramming, or is verified only indirectly through a related test case noted elsewhere in the same module.

2. Module 1 — Customer Registration

FR ID	Requirement (Brief)	Pri.	Use Case	User Story	Diagram(s)	UAT TC
FR-REG-01	Initiate registration entry point	Must	UC-01	US-01	B1, S1	TC-REG-01
FR-REG-02	Mobile number OTP verification	Must	UC-01	US-02	B1, S1	TC-REG-01, 03
FR-REG-03	Login/biometric authentication check	Must	UC-01	US-02	B1	TC-REG-01
FR-REG-04	Device binding capture	Must	UC-01	US-01	B1	TC-REG-01
FR-REG-05	Consent language & capture	Must	UC-01	US-03	B1, S1	TC-REG-04
FR-REG-06	Consent event logging	Must	UC-01	US-03	B1	TC-REG-04
FR-REG-07	CIF profile fetch	Must	UC-01	US-04	S1	TC-REG-01
FR-REG-08	Eligible account discovery	Must	UC-01	US-04	S1	TC-REG-01, 05
FR-REG-09	Success confirmation screen	Must	UC-01	US-05	B1, S1	TC-REG-01
FR-REG-10	Custom VPA handle selection	Should	UC-02	US-06	—	TC-REG-06
FR-REG-11	Link UPI profile to Customer ID	Must	UC-01	US-04	S1	TC-REG-01
FR-REG-12	Registration rejection handling	Must	UC-01	US-02	B1	TC-REG-02

3. Module 2 — Bank Account Linking

FR ID	Requirement (Brief)	Pri.	Use Case	User Story	Diagram(s)	UAT TC
FR-ACC-01	Fetch eligible accounts	Must	UC-03	US-07	B2	TC-ACC-01
FR-ACC-02	Default account designation	Must	UC-05	US-11	B2	TC-ACC-03
FR-ACC-03	Add secondary account	Must	UC-03	US-08	B2	TC-ACC-01
FR-ACC-04	Remove linked account (guarded)	Must	UC-04	US-09, US-10	B2	TC-ACC-02
FR-ACC-05	Change default account	Must	UC-05	US-11	B2	TC-ACC-03
FR-ACC-06	Account nickname display	Should	UC-03	US-07	—	TC-ACC-01
FR-ACC-07	Automatic delinking on closure	Must	UC-03	—	B2	TC-ACC-05
FR-ACC-08	Block ineligible account types	Must	UC-03	US-08	B2	TC-ACC-04

4. Module 3 — UPI PIN Management

FR ID	Requirement (Brief)	Pri.	Use Case	User Story	Diagram(s)	UAT TC
FR-PIN-01	Debit card / Aadhaar-OTP validation	Must	UC-06	US-12, US-13	B3, S2	TC-PIN-01, 02
FR-PIN-02	Mobile OTP validation for PIN	Must	UC-06	US-12	B3, S2	TC-PIN-01
FR-PIN-03	Secure PIN entry pad	Must	UC-06	US-12	B3	TC-PIN-01
FR-PIN-04	On-device PIN encryption	Must	UC-06	US-12	B3, S2	TC-PIN-01
FR-PIN-05	PIN reset without old PIN	Must	UC-07	US-14	B3	TC-PIN-03
FR-PIN-06	PIN change with old PIN validation	Must	UC-08	US-15	—	TC-PIN-04
FR-PIN-07	Lockout after 3 failed attempts	Must	UC-09	US-16	A2	TC-PIN-05
FR-PIN-08	Unlock path after lockout	Should	UC-09	US-17	A2	TC-PIN-06

5. Module 4 — Send Money

FR ID	Requirement (Brief)	Pri.	Use Case	User Story	Diagram(s)	UAT TC
FR-TXN-01	Send via mobile number	Must	UC-10	US-18	B4, S3	TC-TXN-02
FR-TXN-02	Send via UPI ID	Must	UC-11	US-19	B4, S3	TC-TXN-01
FR-TXN-03	Send via QR scan	Must	UC-20	US-32	B4, S6	TC-QR-03
FR-TXN-04	Send via Bank Account + IFSC	Must	UC-12	US-20	B4	TC-TXN-03
FR-TXN-05	Self-transfer between own accounts	Should	UC-13	US-21	B4	TC-TXN-04
FR-TXN-06	Recipient name confirmation	Must	UC-10, UC-11	US-18, US-19	B4, S3	TC-TXN-01, 02
FR-TXN-07	PIN authorisation required	Must	UC-10 to 13	US-18	A1, S3	TC-TXN-01
FR-TXN-08	Real-time balance validation	Must	UC-10 to 13	US-22	A1	TC-TXN-05
FR-TXN-09	Fraud/velocity risk scoring	Must	UC-10 to 13	US-23	A1, S3	TC-TXN-07
FR-TXN-10	Per-transaction/daily limit checks	Must	UC-10 to 13	US-22	A1	TC-TXN-06
FR-TXN-11	Real-time status display	Must	UC-10 to 13	US-18	S3	TC-TXN-01
FR-TXN-12	Save favourite payee	Could	—	—	—	—
FR-TXN-13	Payment note/remark	Could	—	—	—	—
FR-TXN-14	Timeout reconciliation & reversal	Must	UC-14	US-24	A1, S4	TC-TXN-08

6. Module 5 — Collect Requests

FR ID	Requirement (Brief)	Pri.	Use Case	User Story	Diagram(s)	UAT TC
FR-COL-01	Create collect request	Must	UC-15	US-25	B5, S5, A3	TC-COL-01
FR-COL-02	Notify payer in real time	Must	UC-15	US-25	B5, S5	TC-COL-01
FR-COL-03	Approve with PIN authorisation	Must	UC-16	US-26	B5, S5, A3	TC-COL-01
FR-COL-04	Reject with optional reason	Must	UC-17	US-27	B5, A3	TC-COL-02
FR-COL-05	Auto-expire after 24 hours	Must	UC-18	US-28	B5, A3	TC-COL-03
FR-COL-06	Reminders (max 3)	Should	—	US-29	A3	TC-COL-04
FR-COL-07	Cancel pending request	Should	—	—	—	—
FR-COL-08	Status display (all states)	Must	UC-15 to 18	—	—	—

7. Module 6 — QR Payments

FR ID	Requirement (Brief)	Pri.	Use Case	User Story	Diagram(s)	UAT TC
FR-QR-01	Static QR generation	Must	UC-19	US-30	A4	TC-QR-01
FR-QR-02	Dynamic QR generation	Must	UC-19	US-31	A4	TC-QR-02
FR-QR-03	Scan third-party/merchant QR	Must	UC-20	US-32	A4, S6	TC-QR-03
FR-QR-04	Merchant registry validation	Must	UC-20	US-33	A4	TC-QR-03, 04
FR-QR-05	Save/download QR image	Could	UC-19	US-34	—	—
FR-QR-06	Blacklist enforcement	Must	UC-20	US-33	A4	TC-QR-04

8. Module 7 — Balance Check

FR ID	Requirement (Brief)	Pri.	Use Case	User Story	Diagram(s)	UAT TC
FR-BAL-01	PIN-authenticated balance enquiry	Must	UC-21	US-35, US-36	—	TC-BAL-01, 03
FR-BAL-02	Account nickname on balance screen	Should	UC-21	US-35	—	TC-BAL-01
FR-BAL-03	Daily enquiry cap enforcement	Must	UC-21	US-37	—	TC-BAL-02

9. Module 8 — Transaction History

FR ID	Requirement (Brief)	Pri.	Use Case	User Story	Diagram(s)	UAT TC
FR-HIS-01	Default 6-month chronological list	Must	UC-22	US-38	—	TC-HIS-01
FR-HIS-02	Multi-criteria filtering	Must	UC-22	US-39	—	TC-HIS-02
FR-HIS-03	PDF statement export	Should	UC-23	US-41	—	TC-HIS-03
FR-HIS-04	Transaction detail drill-down	Must	UC-22	US-40	—	TC-HIS-04
FR-HIS-05	10-year retention & search	Must	UC-22	US-38	—	TC-HIS-01

10. Module 9 — Notifications

FR ID	Requirement (Brief)	Pri.	Use Case	User Story	Diagram(s)	UAT TC
FR-NOT-01	Payment success/failure notification	Must	UC-24	US-42	A5	TC-NOT-01
FR-NOT-02	Failed transaction notification	Must	UC-24	US-42	A5	TC-NOT-01
FR-NOT-03	Collect request notifications	Must	UC-24	US-43	A5	TC-NOT-05
FR-NOT-04	Refund/chargeback notification	Must	UC-24	—	A5	—
FR-NOT-05	Registration/PIN confirmation	Must	UC-24	—	A5	—
FR-NOT-06	Security alert notification	Must	UC-24	US-44	A5	TC-NOT-02
FR-NOT-07	Marketing opt-in/out preference	Should	UC-25	US-45	—	TC-NOT-03

11. Module 10 — Dispute Management

FR ID	Requirement (Brief)	Pri.	Use Case	User Story	Diagram(s)	UAT TC
FR-DIS-01	Raise dispute within 180 days	Must	UC-26	US-47	B6, S7, A6	TC-DIS-01, 02
FR-DIS-02	Dispute category selection	Must	UC-26	US-47	B6	TC-DIS-01
FR-DIS-03	Internal reference number generation	Must	UC-26	US-47	S7	TC-DIS-01
FR-DIS-04	NPCI reference number capture	Must	UC-26	US-49	S7, A6	TC-DIS-04
FR-DIS-05	Dispute status tracking	Must	UC-27	US-49	—	TC-DIS-04
FR-DIS-06	Refund status & credit date display	Must	UC-27	US-51	—	TC-DIS-05
FR-DIS-07	Automated TAT escalation	Should	UC-29	US-52	A6	TC-DIS-06
FR-DIS-08	Internal investigation notes (Ops)	Must	UC-28	US-50	—	TC-DIS-03

12. Non-Functional Requirement Traceability

Non-functional requirements are verified primarily through the dedicated NFR test cases in the UAT Scenarios deliverable rather than through Use Cases or User Stories, though several correspond to explicit design decisions noted in the Activity Diagrams and API Integration Specification.

NFR ID	Requirement (Brief)	Verified By	UAT TC
NFR-SEC-02	PIN never stored/logged in plaintext	PSP Switch API design (encrypted_pin_block); FRS PIN module	TC-NFR-03
NFR-SEC-05	Pre-go-live penetration testing	Security test cycle (external to UAT)	TC-NFR-04
NFR-PERF-01	Transaction latency $\leq 8s$ average	API timeout targets in API Integration Spec	TC-NFR-01
NFR-PERF-02	500 TPS peak load capacity	Load test plan	TC-NFR-02
NFR-AVL-02	PSP switch/API gateway failover	Integration architecture (FRS Section 12)	TC-NFR-05
NFR-AVL-03	DR — RTO $\leq 4h$, RPO $\leq 15min$	DR test plan	TC-NFR-06
NFR-USE-02	Core flows ≤ 4 screens	Wireframes (Send Money, Scan & Pay flows)	TC-NFR-07
NFR-CMP-03	India-only data hosting	Infrastructure/hosting configuration review	TC-NFR-08

13. Coverage Summary

Module	Total FRs	Traced to Use Case	Traced to User Story	Traced to Diagram	Traced to UAT TC
Customer Registration	12	12	12	11	12
Bank Account Linking	8	8	7	7	7
UPI PIN Management	8	8	8	6	8
Send Money	14	12	12	12	12
Collect Requests	8	6	6	6	5
QR Payments	6	6	6	5	5
Balance Check	3	3	3	0	3
Transaction History	5	5	5	0	5
Notifications	7	7	5	7	3
Dispute Management	8	8	8	6	7
TOTAL	79	75	72	60	67

Gaps are concentrated in Could Have items deferred from detailed design (FR-TXN-12/13, FR-COL-07, FR-QR-05) and a small number of Should Have notification/status requirements verified only indirectly (FR-NOT-04/05, FR-COL-08, FR-DIS-05/06 partial). None of the gaps fall on a Must Have requirement, so no Must Have item is without both a design artefact and a test case. These gaps should be closed opportunistically as the backlog is refined, or formally accepted as residual scope via the change control process defined in the BRD.